7.

$$\begin{cases} x \cdot \sin 10° - y \cdot \cos 10° = -\dfrac{1}{\sin 50°} & \times(-\sin 50°) \\ x \cdot \sin 50° + y \cdot \cos 50° = \dfrac{1}{\sin 10°} & \times \sin 10° \end{cases}$$

$$\begin{cases} -x \cdot \sin 10° \cdot \sin 50° + y \cdot \cos 10° \cdot \sin 50° = 1 \\ x \cdot \sin 10° \cdot \sin 50° + y \cdot \sin 10° \cdot \cos 50° = 1 \end{cases} \oplus$$

$$y \cdot \cos 10° \cdot \sin 50° + y \cdot \sin 10° \cdot \cos 50° = 2$$

$$y \cdot (\cos 10° \cdot \sin 50° + \sin 10° \cdot \cos 50°) = 2$$

$\sin a = 50°, b = 10°$

$\sin a \cdot \cos b + \sin b \cdot \cos a$

$\sin(a+b)$

$$y \cdot \sin(\underbrace{50° + 10°}_{60°}) = 2 \Rightarrow y \cdot \frac{\sqrt{3}}{2} = 2 \Rightarrow y = \frac{4}{\sqrt{3}}$$

$$\therefore y = \frac{4\sqrt{3}}{3}$$

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